



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL APPLICATIONS [SETA]

FULL MARKS: 100
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 2 printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

Question 1

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only)

- (i) Euro II is an example of permissible emission norms by [1]
(a) Industry (b) Vehicles
(c) Thermal power plants (d) Brick kilns
- (ii) The process of burning municipal solid wastes under suitable temperature is [1]
(a) Landfill (b) Composting
(c) Incineration (d) Ignition
- (iii) Study the picture given below and identify the soil conservation strategy. [1]



- (a) Contour bunds (b) Tree break
(c) Terrace farming (d) Forested area

- (iv) Band aid is an example of [1]
 (a) Domestic waste (b) Hazardous waste
 (c) Biomedical waste (d) E waste
- (v) Identify the incorrect statement: [1]
 (a) Agro forestry adds more pressure to forests
 (b) Agro forestry demonstrates proper utilization of farm resources
 (c) In agro forestry, trees help in checking soil erosion by wind.
 (d) Agro forestry is a self sustaining system.
- (vi) Eris is temple ponds found in Tamil Nadu where traditional structures [1]
 are built to harvest rainwater. Select the appropriate statement
 emphasizing on the advantages of Eris.
 (i) Water is used for bathing and religious purposes.
 (ii) Water overflows causing floods.
 (iii) Water is used for washing purposes.
 (iv) Water is used for irrigation of croplands.
 (a) i and ii (b) i, ii, iv (c) i and iii (d) i only
- (vii) Assertion: Delhi is one of the most polluted cities in the world. [1]
 Reason: Delhi government has introduced CNG for public transport in
 Delhi.
 (a) Both assertion and reason are true and reason is the correct
 explanation of assertion
 (b) Both assertion and reason are true and reason is not the correct
 explanation of assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.
- (viii) Assertion: Forest land in our country is 21% of the country's total land. [1]
 Reason: Industrializations acquire a lot of forest land and use up a lot of
 timber.
 (a) Both assertion and reason are true and reason is the correct
 explanation of assertion
 (b) Both assertion and reason are true and reason is not the correct
 explanation of assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.
- (ix) In a semi arid region of Rajasthan, farmers face frequent wind erosion of [1]
 soil and loss of topsoil fertility. To combat this, the government promotes
 the planting of shelter belts.
 In what direction should shelter belts be planted to be most effective?
 (a) Parallel
 (b) Random pattern
 (c) Along riverbank only
 (d) Perpendicular to the prevailing wind direction

- (x) Why is biogas considered a better alternative to fuel wood cooking? [1]
 (a) Produces more smoke
 (b) More expensive
 (c) Renewable and produces less indoor pollution
 (d) Requires cutting of trees
- (xi) A farmer lets his cattle graze continuously on the same pasture without rest. What is the likely outcome? [1]
 (a) Increase in soil nutrients
 (b) Regeneration of plant cover
 (c) Depletion of grass and exposed soil
 (d) Enhanced forest growth
- (xii) Among the statements given below select the correct ones that describe the concept of mulching. [1]
 (i) Water conservation technique
 (ii) Boulders are used for the process
 (iii) Planting of leguminous plants like peas and grass like alpha alpha are included in the process
 (iv) Adds organic matter to water
 (a) i and iv (b) ii and iii (c) i, ii and iv (d) iii only
- (xiii) Which of the following statements best describes the relationship between watershed management and flood control? [1]
 (a) Watershed management practices like reforestation and check dam help control the flow of water and reduce flood risks
 (b) Watershed management has no effect on flood control
 (c) Watershed management increases flood risks by blocking water flow
 (d) Watershed management helps to decrease water storage in rivers, leading to more flooding.
- (xiv) Which harmful substances can be found in E wastes? [1]
 (a) Iron & calcium (b) Nitrogen & oxygen
 (c) Water & carbondioxide (d) Lead & cadmium
- (xv) Select the correct sequence of phenomenon for ESP. [1]
 (i) Collection of charged particles
 (ii) Charging of particles
 (iii) Release of clean air
 (iv) Ionization
 (a) i, iii, ii, iv (b) iii, i, ii, iv (c) ii, iv, iii, i (d) iv, ii, i, iii
- (xvi) _____ are law suit to protect public interest [1]
 (a) PIL (b) PLI (c) PCB (d) PUC

- (xvii) A hybrid vehicle uses [1]
 (a) Two or more distinct power sources to move the vehicle
 (b) only coal to move the vehicle
 (c) nitrogen gas to move the vehicle.
 (d) Biogas to move the vehicle
- (xviii) Select the correct pair: [1]
 (a) SPM – water droplets
 (b) Bag filter – dust particles
 (c) Secondary treatment- chlorine
 (d) Toxic waste- husks
- (xix) Subsidies are: [1]
 (a) Taxes imposed by the Government
 (b) Surcharges on the non ecofriendly products
 (c) Financial aid given by a government
 (d) Fine imposed by Government on act committed against law
- (xx) The source of electricity generated by large dam is [1]
 (a) Tidal wave
 (b) Stored falling water
 (c) Turbine
 (d) Generator

Question 2

- (i) Mention 2 alternatives to timber in the building industry. [2]
 (ii) Why is there a ban on single use plastics? [2]
 (iii) List any three impacts of siltation in waterways. [3]
 (iv) The automobile industry plays a crucial role in a country's economy, but also contributes to environmental pollution. To address this, the EU introduced Euro norms in 1990s which became stricter over time. [3]
 Why did India not adopt Euro norms?

Question 3

- (i) What are HOVs? Are they helpful or harmful for the environment? [2]
 Justify your answer.
 (ii) Write two benefits of kunds. [2]
 (iii) Identify three sources of indoor air pollution. [3]

(iv)

[3]



Identify the type of waste. Why is it harmful? Who are the major contributors of such wastes?

SECTION B (60 MARKS)
(Attempt any four questions from this section)

Question 4

(i) Study the picture and answer the questions that follows:

[5]



(a) Where is it situated?

(b) List any four sustainable practices adopted by the people of Auroville.

(ii) Mention any five steps that can be taken to prevent deforestation. [5]

(iii) In the last two decades, several Indian cities have seen rapid urban development. Cities like Bangalore, Hyderabad and Gurgaon have lost significant portion of their wetlands to construction, road building and real estate expansion. For example Bengaluru had over 200 wetlands in 1970s, but today fewer than 30 functional wetlands remain. State 2 reasons and 3 consequences for the decline of wetlands in Indian cities. [5]

Question 5

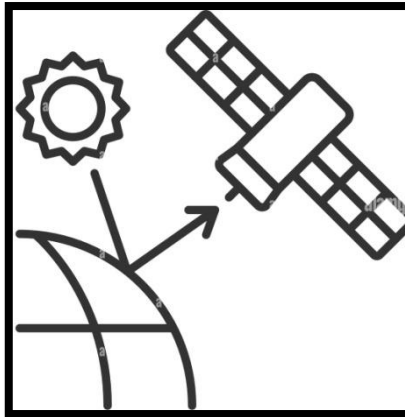
- (i) Pesticides are chemicals sprayed on plants or added to soil to eliminate harmful plant pests. They help in increasing crop yield by killing pests, but have harmful effect on environment and humans. Highlight the limitations of using chemical pesticides. [5]
- (ii) What is social forestry? How does it benefit common people? [5]
- (iii) 'Bubble Theory' is one of the strategies reducing air pollution. Why is it known as Bubble Theory? Give an account of the theory. [5]

Question 6

- (i) Enumerate the role of local people in JFM. [5]
- (ii) List any five problems associated with large dams. [5]
- (iii) Explain the working of Cyclone separator. [5]

Question 7

- (i) What is a dry compost toilet? Give any four advantages of using dry compost toilets. [5]
- (ii) Write a short note on Curitiba's traffic management. [5]
- (iii) [5]



- (a) Name the device shown in the picture.
- (b) Give two reasons why it is considered a good tool.
- (c) Mention two applications of it.

Question 8

- (i) Water is the elixir of life, a very important natural resource. Your teacher wants you to prepare a project plan for assessment. "How to save water"
 - (a) Write any three ways that you will suggest to bring awareness in your neighbourhood.
 - (b) Recommend any two methods to prevent further depletion of the underground water table.
- (ii) What is TTZ? Mention the steps taken by Mathura Oil Refinery to protect the TTZ. [5]
- (iii) Why the secondary treatment in a water treatment plant is called biological treatment? What happens in tertiary treatment? After which [5]

stage is it safe to discharge the treated water in water bodies and why?

Question 9

- (i) Delhi is the city with the highest pollution level in India. Explain any five measures taken by the concerned authorities to reduce air pollution in the city. [5]**
- (ii) You have learnt about various water harvesting techniques. Name and describe the one you would want to implement in your locality. [5]**
- (iii) Decentralised wastewater treatment offers a flexible, cost effective and environment friendly alternative to traditional centralized systems, making it particularly suitable for diverse urban and rural contexts. Justify the statement. [5]**